$$
\begin{aligned}
C_{\phi B}^{(22)} \to \frac{1}{16\,\pi^2}\Bigg(&\frac{1}{36}\sum_p g_1^4\,LF_{3,0}\left[m_{\tilde{d}}^{p}\right]-\frac{1}{36}\sum_p g_1^4\,LF_{4,-1}\left[m_{\tilde{d}}^{p}\right]- \\
&\frac{1}{6}\,g_1^2\,\overline{y_d}^{pr}\,y_d^{pr}\,LF_{3,0}\left[m_{\tilde{d}}^{r}\right]+\frac{1}{6}\,g_1^2\,\overline{y_d}^{pr}\,y_d^{pr}\,LF_{4,-1}\left[m_{\tilde{d}}^{r}\right]+\frac{1}{4}\sum_p g_1^4\,LF_{3,0}\left[m_{\tilde{e}}^{p}\right]- \\
&\frac{1}{4}\sum_p g_1^4\,LF_{4,-1}\left[m_{\tilde{e}}^{p}\right]-\frac{1}{2}\,g_1^2\,\overline{y_e}^{pr}\,y_e^{pr}\,LF_{3,0}\left[m_{\tilde{e}}^{r}\right]+\frac{1}{2}\,g_1^2\,\overline{y_e}^{pr}\,y_e^{pr}\,LF_{4,-1}\left[m_{\tilde{e}}^{r}\right]- \\
&\frac{1}{16}\,g_1^2\left(2\,\overline{y_e}^{pr}\,y_e^{pr}+\sum_p g_1^2\right)LF_{3,0}\left[m_{\tilde{l}}^{p}\right]+\frac{1}{16}\left(2\,g_1^2\,\overline{y_e}^{pr}\,y_e^{pr}+\sum_p g_1^4\right)LF_{4,-1}\left[m_{\tilde{l}}^{p}\right]+ \\
&\frac{1}{144}\left(-6\,g_1^2\,\overline{y_d}^{pr}\,y_d^{pr}+\sum_p g_1^4\right)LF_{3,0}\left[m_{\tilde{q}}^{p}\right]+\frac{1}{144}\left(6\,g_1^2\,\overline{y_d}^{pr}\,y_d^{pr}-\sum_p g_1^4\right)LF_{4,-1}\left[m_{\tilde{q}}^{p}\right]- \\
&\frac{2}{9}\sum_p g_1^4\,LF_{3,0}\left[m_{\tilde{u}}^{p}\right]+\frac{2}{9}\sum_p g_1^4\,LF_{4,-1}\left[m_{\tilde{u}}^{p}\right]-\frac{1}{6}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{2,2,0}\left[m_{\tilde{d}}^{r},\,m_{\tilde{q}}^{p}\right]- \\
&\frac{1}{6}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{3,1,0}\left[m_{\tilde{d}}^{r},\,m_{\tilde{q}}^{p}\right]+\frac{1}{6}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{3,2,-1}\left[m_{\tilde{d}}^{r},\,m_{\tilde{q}}^{p}\right]+ \\
&\frac{1}{6}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{4,1,-1}\left[m_{\tilde{d}}^{r},\,m_{\tilde{q}}^{p}\right]-\frac{1}{2}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{2,2,0}\left[m_{\tilde{e}}^{r},\,m_{\tilde{l}}^{p}\right]- \\
&\frac{1}{2}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{3,1,0}\left[m_{\tilde{e}}^{r},\,m_{\tilde{l}}^{p}\right]+\frac{1}{2}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{3,2,-1}\left[m_{\tilde{e}}^{r},\,m_{\tilde{l}}^{p}\right]+ \\
&\frac{1}{2}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{4,1,-1}\left[m_{\tilde{e}}^{r},\,m_{\tilde{l}}^{p}\right]+\frac{1}{2}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{3,2,-1}\left[m_{\tilde{l}}^{p},\,m_{\tilde{e}}^{r}\right]- \\
&\frac{1}{4}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{4,1,-1}\left[m_{\tilde{l}}^{p},\,m_{\tilde{e}}^{r}\right]+\frac{1}{4}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{5,1,-2}\left[m_{\tilde{l}}^{p},\,m_{\tilde{e}}^{r}\right]+ \\
&\frac{1}{3}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{3,1,0}\left[m_{\tilde{q}}^{p},\,m_{\tilde{d}}^{r}\right]+\frac{1}{6}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{3,2,-1}\left[m_{\tilde{q}}^{p},\,m_{\tilde{d}}^{r}\right]- \\
&\frac{13}{12}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{4,1,-1}\left[m_{\tilde{q}}^{p},\,m_{\tilde{d}}^{r}\right]+\frac{3}{4}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{5,1,-2}\left[m_{\tilde{q}}^{p},\,m_{\tilde{d}}^{r}\right]- \\
&\frac{1}{24}\,g_1^2\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{2,2,0}\left[m_{\tilde{q}}^{p},\,m_{\tilde{u}}^{r}\right]-\frac{1}{24}\,g_1^2\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{3,1,0}\left[m_{\tilde{q}}^{p},\,m_{\tilde{u}}^{r}\right]+ \\
&\frac{1}{24}\,g_1^2\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{3,2,-1}\left[m_{\tilde{q}}^{p},\,m_{\tilde{u}}^{r}\right]+\frac{1}{24}\,g_1^2\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{4,1,-1}\left[m_{\tilde{q}}^{p},\,m_{\tilde{u}}^{r}\right]- \\
&\frac{7}{24}\,g_1^2\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{3,1,0}\left[m_{\tilde{u}}^{r},\,m_{\tilde{q}}^{p}\right]+\frac{1}{24}\,g_1^2\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{3,2,-1}\left[m_{\tilde{u}}^{r},\,m_{\tilde{q}}^{p}\right]- \\
&\frac{11}{24}\,g_1^2\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{4,1,-1}\left[m_{\tilde{u}}^{r},\,m_{\tilde{q}}^{p}\right]+\frac{3}{4}\,g_1^2\,\tilde{\mu}^2\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{5,1,-2}\left[m_{\tilde{u}}^{r},\,m_{\tilde{q}}^{p}\right]- \\
&\frac{1}{8}\,g_1^4\,LF_{3,1,-1}\left[\tilde{\mu},\,m_1\right]+\frac{3}{8}\,g_1^4\,LF_{4,1,-2}\left[\tilde{\mu},\,m_1\right]-\frac{1}{4}\,g_1^4\,LF_{5,1,-3}\left[\tilde{\mu},\,m_1\right]- \\
&\frac{3}{8}\,g_1^2\,g_2^2\,LF_{3,1,-1}\left[\tilde{\mu},\,m_2\right]+\frac{9}{8}\,g_1^2\,g_2^2\,LF_{4,1,-2}\left[\tilde{\mu},\,m_2\right]-\frac{3}{4}\,g_1^2\,g_2^2\,LF_{5,1,-3}\left[\tilde{\mu},\,m_2\right]\Bigg)
\end{aligned}
$$